

The Founding Fathers of Sociology: Francis Galton, Adolphe Quetelet, and Charles Booth Or What Do People You Probably Never Heard of Have to Do with the Foundations of Sociology?

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ABSTRACT

The theme of the 2008 meeting of the Association for Applied and Clinical Sociology was “Engaging Sociology: Applied Sociology’s Past, Present, and Promise.” The theme suggests that applied sociology has a past that is different from the parent discipline of sociology, and given how the history of sociology has come to be taught and remembered, that is an understandable suggestion. The argument of this paper, however, is that the discipline of sociology itself—as it is actually practiced today—originated mainly in applied work, in the work of nineteenth century social reformers whose contributions to the field have been largely forgotten, people such as Francis Galton, Adolphe Quetelet, and Charles Booth. These, I argue, are the Founding Fathers of the discipline as I have practiced it for the past thirty-five years.

AT THE MOST GENERAL LEVEL, the topic of this paper involves the relationship between basic and applied science. Most discussions of this relationship, be they in the natural or social sciences, posit the development of the basic scientific discipline—its theories, concepts, methods, and corpus of findings—as the necessary, indeed logically necessary, first step, and then the application of those theories, methods, and findings to problems of direct practical importance as a derivative activity, as an inferior and much less glamorous second step. The idea is that the theoretical and methodological tools are developed first; only after they are developed does one look around to see what they can be used for. Scientific accolades, in turn, fall to the inventors of tools and generally not to those who apply them.

The thesis of this paper is that the development of sociology as an intellectual and academic discipline followed an exactly opposite pattern. Sociology as it is actually practiced was “invented” by practical people seeking solutions to the pressing social problems of their day. Sociology, I argue, was an applied enterprise long before it was developed as a “basic” scientific discipline.

We have all been taught, and have all probably taught our own students, that the origins of sociology lie in the theoretical speculations and conceptual development of figures such as the following:

Auguste Comte (1798–1857), credited with inventing the term *sociology* and sometimes described as either the “first Western sociologist” or the “founder of functionalism.”¹ Comte is famous for suggesting that society has gone through three phases: the theological, the metaphysical, and the scientific; and for his observation that “If it is true that every theory must be based upon observed facts, it is equally true that facts cannot be observed without the guidance of some theories. Without

such guidance, our facts would be desultory and fruitless; we could not retain them: for the most part we could not even perceive them" (Comte 1853).

Emile Durkheim (1858–1917), often called "the father of sociology" and often credited with making sociology a "science" by insisting that social facts can only be explained by social facts, a radical proposition at the time because it seemed to deny the importance, or even the very possibility, of volition and individual moral duty. It is true, unequivocally, that Durkheim's many scholarly works and his founding and editing of the first journal of sociology, *L'Année Sociologique*, helped establish sociology within academia as an accepted social science.

Max Weber (1864–1920), the youngest of the founders, who, although considered along with Karl Marx and Émile Durkheim as one of the founders of modern sociology, thought of himself and was known by his contemporaries primarily as a historian and an economist. Today, we tend to think of Weber as an abstract thinker, a theoretician, but he was intimately engaged in practical affairs and German politics: for example, he was a member of the German delegation during the peace negotiations in Versailles and a member of the committee that drafted the disastrous Weimar constitution that paved the way for Hitler's ascension to power a decade or so later.²

Among other generally recognized "founders" would also be included Karl Marx (1818–1883), who, like Weber, was more economist and historian (and rabble-rouser) than sociologist, but whose ideas about society and its development continue to resonate in the discipline; and the notorious Herbert Spencer (1820–1903), remembered as a sociological theorist of the Victorian era, coined the term "survival of the fittest" (note: not Charles Darwin, a contemporary) and who wrote at great length on the evolutionary progression of society from simple, undifferentiated homogeneity to complex, differentiated, heterogeneity, and more specifically, as he put it, the progression from militant to industrial society. (A little-known historical irony is that the graves of both Karl Marx and Herbert Spencer are in London's Highgate Cemetery, each gravesite easily visible from the other.)

Venerated as these historical figures are, one might profitably ask: Just what aspects or features of their work survive in contemporary sociology? Marx probably remains the most influential of the batch, although Marxian sociological themes have fallen out of favor in recent years. No one writes about suicide without a tip of the hat to Durkheim, but do the *Rules of Sociological Method*, the *Elementary Forms of the Religious Life*, or even *The Division of Labor in Society* still inform much current sociological research or practice? Weber, of course, lives on in studies of bureaucracy and in studies of "class, status and power," and his *Protestant Ethic and the Spirit of Capitalism* is a hardy perennial in undergraduate social theory classes, but beyond that, what? As for Comte and Spencer, nothing further need be said.

Consider, in sharp contrast, Sir Francis Galton (1822–1911), a half-cousin of Charles Darwin and a contemporary of Marx, Weber, Durkheim, and most of the other alleged Founding Fathers. Galton was a true genius, a polymath, who would have been described at various points in his career as an anthropologist, sociologist, explorer, geographer, meteorologist, geneticist, psychometrician, criminologist, and statistician, among others.³

Tellingly, Robert Nisbet's (1966) much-quoted and revered history of sociology, *The Sociological Tradition*, something still read by graduate students to prepare for their comprehensive examinations in sociological theory, does not contain a single reference to Galton anywhere in its ponderous 350 pages, although there are references in the book to such important sociological personages as Georg Maurer, Adam Ferguson, Francis Bradley, and a large number of others long since forgotten. And most sociologists have evidently never heard of the man. So let me summarize a few of Galton's intellectual contributions.

- According to Karl Pearson, he of the Pearson product-moment correlation coefficient, Galton invented the concept of statistical correlation. (Pearson 1914)

- Galton was the first to apply statistical methods to the study of human differences and the inheritance of intelligence.
- Galton introduced the use of questionnaires and surveys for collecting data on human communities, which he employed in his genealogical and biographical works and for anthropometric studies. In an important sense, the man invented the modern survey questionnaire and other features of the modern survey.
- He coined the phrase “nature vs. nurture” and was the first person to study genetic inheritance vs. environmental conditioning and socialization as determinants of human characteristics.
- He is also considered the founder of psychometrics (the science of measuring mental faculties and psychological and social-psychological properties).
- In the late 1860s, Galton conceived the concept of the standard deviation.
- Galton also invented ordinary least squares regression and the use of the regression line; he was also the first to describe and explain the common phenomenon of regression toward the mean.
- In the 1870s and 1880s he pioneered the use of the normal distribution to fit histograms of actual tabulated data. He also discovered the properties of the bivariate normal distribution and its relationship to regression analysis.

In other words, contemporary sociology as it is actually practiced—and by this I mean the stuff that I and my colleagues do every day—is literally inconceivable without reference to the techniques, concepts, and theories pioneered by Sir Francis Galton. I go days, weeks, even months at a time without reference to anomie, class consciousness, or Max Weber, but the ghost of Sir Francis Galton hovers over my desk every working minute. Surely, these accomplishments would seem to rate at least some mention in a history of *The Sociological Tradition*, but no, apparently not.

It is worth asking about the motivations that drove Galton to this dizzying string of discoveries and breakthroughs. Clearly, those motivations were practical, not theoretical. Galton’s passion was the statistical study of variations in human ability. The fundamental observation that drove his scientific work was that some people possessed talent in great abundance, and others, barely at all. What accounts for that variation in human ability and could anything be done about it? A more practical concern is difficult to conjure! As James Newman (1956) put it in his monumental history of mathematics, Galton was “essentially a social reformer,” “imbued with a strong sense of social responsibility and of mission.”

By modern sociological standards, of course, Galton ended up on the wrong side of the nature-nurture issue. While we believe (as much as an article of faith as an evidentiary conclusion) that nurture trumps nature, Galton’s quantitative, ethnographic, and biological studies led him more and more to the conclusion that ability was inherited, not a product of environment or socialization. As he expressed it in his work, *Hereditary Genius* (1869):

I have no patience with the hypothesis occasionally expressed, and often implied, especially in tales written to teach children to be good, that babies are born pretty much alike, and that the sole agencies in creating differences between boy and boy, and man and man, are steady application and moral effort. It is in the most unqualified manner that I object to pretensions of natural equality. The experiences of the nursery, the school, the University, and of professional careers, are a chain of proofs to the contrary.

From these ideas Galton found himself involved in the field of hereditary improvement, which he called eugenics (another term that he coined). Today, we associate “eugenics” with racism, class prejudice, and the discredited thinking of evil Nazi biologists intent on breeding a master race, yet this is a far cry from Galton’s own thoughts and intentions. If the lot of humankind could only be

improved through genetics, and not through education or moral instruction, he reasoned, then the practical question must be raised, How might one breed feeble-mindedness, poverty, and misery out of the population? Today's sensibilities are offended by even posing the question, but to the sensibilities of 19th century Victorians, it was as obvious a question as the question how one might breed a population of superior beef cattle or more handsome marigolds. In fact, to 19th century sensibilities, these are the exact same questions.

The life of Adolphe Quetelet (1796–1894) overlapped that of Galton and most of the alleged founding fathers as well.⁴ Quetelet was one of the most influential social scientists of the nineteenth century. He pioneered a field of investigation that came to be known as “political arithmetic” or, in another rendition, “moral statistics.” Like Galton, he was a pioneer in the use of statistical reasoning to study social phenomena and as such, he profoundly influenced the course of European and eventually American social science. Florence Nightingale, whose appreciation of statistics and statistical inference was very much more refined than that of today's average student and who in 1858 became the first woman elected to the Royal Statistical Society, considered him something like the social science equivalent of Isaac Newton.

Quetelet is rarely even mentioned in modern-day histories of the development of social science, but his efforts to apply probability theory and statistical analysis to data on crime, suicide, and anthropometry were pioneering efforts whose progeny include criminology in its contemporary quantitative form. Nisbet's (1966) history of sociology contains a single and very condescending mention of Quetelet, who is identified as one of the “political arithmeticians” who “compiled quantities of social statistics, pointing to correlations or ‘patterns,’ and preening themselves on the quantitative exactitude of their handiwork.” This snide comment notwithstanding, Quetelet was a venerated figure throughout the nineteenth century among empirical researchers in Victorian England and on the Continent (Cole 1972).

Quetelet became interested in statistics and sociology around 1830. He was obsessed with the concept of the “Average Man” because he was convinced that probability, randomness, and error were principle determinants of the course of human affairs—that is, an average and then deviations around that average. His obsession led him to develop statistical profiles of the average physical, intellectual, and social features of various populations. From this he attempted to identify the underlying regularities for both normal and abnormal behavior. Much of his work reads like a 19th century precursor to *the Statistical Abstract of the United States*.

Quetelet thought of his “averages” as properties of particular people, groups, or races, that is, as things waiting to be discovered and not just abstract concepts. What he called his “Average Man” was nothing more or less than the central tendency of a distribution of some human trait with deviations around the central tendency that looked like a normal bell-shaped curve. Quetelet was the first to notice this distributional feature of many human properties. In his later work, he even described the Average Man as an ideal type that nature “shoots at” with aiming errors that produce normally distributed deviations. There is no doubt that this concept had a powerful influence on the thinking of Sir Francis Galton, among others.

What Nisbet (1966) described as “preening on the quantitative exactitude of statistical handiwork” was in fact a near-obsession with accurate measurements of human characteristics. Quetelet believed that if measurements were sufficiently accurate, then observed sample values could be used to infer properties of entire populations. He did not quite grasp our modern concept of statistical inference, but as can be plainly seen, he came very close.

Quetelet applied these concepts to statistics on crime, mortality, weight, disease, intelligence, occupations, levels of education, religious affiliation, economics, and other phenomena. His emphasis on accuracy in measurement led him to devise many improvements in census taking and data

gathering. In a pamphlet on crime issued in 1831, he stated what he considered the two central principles for the study of mankind at large:

First: Causes are proportional to the effects they produce. If a man can lift twice as much as another man, he is twice as strong as that man. Study of man's moral qualities [and social characteristics] is possible only if this principle can be applied to them also. [Note: Here Quetelet is calling explicitly for interval level measurement as a necessary condition for a science of "man's moral qualities. . . ." How else could one decide whether the observed effects are "proportional" to the alleged causes?]

Second: Exact conclusions are possible only with large numbers. "The greater the number of individuals, the more the influence of the individual will be effaced, being replaced by the series of general facts that depend on the general causes according to which society exists and maintains itself."⁵

In other words, Quetelet intuited what we know today as the law of large numbers and understood, at least intuitively, the effects of increasing sample sizes on what we now call "the standard error of the estimate."

Like the other political arithmeticians and moral statisticians, Quetelet's scientific work was motivated by a strong vision of social reform, not by the thought of developing Grand Theories of Society. He was interested in crime because he wanted to end it; in mortality because he wanted to delay it; in education because he wanted to maximize it. To the extent that people like Galton and Quetelet had any academic agenda at all over and beyond their essentially reformist interests, it was to demonstrate that a true science of society was indeed possible and to show what such a science would look like. And in their viewpoint, the very essence of that science would be detailed, empirical, quantitative descriptions of the problems of the human condition (Wright 1988). The pressing need was to develop the methodologies by which these descriptions could be generated, assuredly not to develop a body of propositions to be tested by methods that would be invented later. And it is not unfair to say that this vision of "a true science of society" finds more and deeper expression in any given issue of the *American Sociological Review* than anything one will find in Durkheim's *Rules of Sociological Method* or Weber's famous essay on "Politics as a Vocation."

Most sociologists have probably at least heard of the third person mentioned in the subtitle of this paper, Charles Booth (1840–1916), another figure whose life overlaps with all the other "Founding Father" scholars so far discussed. Like Galton, Booth also goes unmentioned in Nisbet's history of sociology, which is little short of astonishing since "many people trace the origins of modern survey research to Charles Booth, who produced a landmark study titled *Life and Labor of the People of London*," published in seventeen volumes (!) over the period 1889–1903 (Groves et al. 2004:4).

Given the importance of survey research to the contemporary sociological enterprise, one might think that the person acknowledged as the inventor of the modern version of the method might bear at least a passing mention in any history of the discipline, but, again, no. Perhaps Nisbet felt he needed to conserve space for such sociological eminences as Richard Price, Frederick Seebohm, and William Cobbett, all of whose work receives some attention.

If we accept the conventional idea that Booth's London surveys were the first "true" sample surveys of the modern form, then we must again acknowledge the primary role of applied or policy issues in the conception and maturation of the discipline. Madge's (1962:536) comments on Booth's work and on the "social survey movement" that Booth initiated are insightful and worth quoting at length. "The social survey movement," he says, ". . . was generally concerned with the type of sociology needed to establish the incidence of social evils, primarily with poverty, with the purpose of arousing public opinion into taking practical measures to deal with such evils. One cannot disagree with the verdict . . . that the social survey is not concerned with evolving any comprehensive sociological theory."

What were Booth's concerns? Booth had extensively reviewed the existing statistical data on poverty in England by analyzing census returns. He concluded that the existing data were misleading, incomplete, and highly unsatisfactory. He subsequently gained public recognition—some would say notoriety—by criticizing then current estimates of the extent of poverty in London. The unprecedented scale of the British poverty problem was the frequent subject of sensationalized press reporting and, coupled with proletarian agitation elsewhere in Europe, these reports must surely have provoked alarm in middle and upper class Victorians. Booth recognized at once that accurate descriptive data would be essential to combat alarmist conjecture in the press. Who among us has not shared this motivation in our own applied work?

Specifically, H. M. Hyndman, the leader of Britain's first socialist party, had stated in a speech in 1885 that 25 percent of Londoners lived in abject poverty. For several years before this, Booth had been investigating poverty in the East End of London with a research team that included his cousin Beatrice Potter, she of children's book fame. This work showed an incidence of pauperism closer to 35 percent than 25 percent, and Booth considered this difference sufficiently large to engage in a public debate. In the course of this debate, the question of what would qualify as "abject poverty" was brought up, and Booth argued, based largely on the cost of living at the time, that anything less than 10–20 shillings per week for a family of four or five people would constitute abject poverty. It will be noticed at once, then, that Booth essentially invented the idea of a "poverty line" based on incomes and the cost of living and adjusted for family size.

The argument with Hyndman caused Booth to expand his research to the whole of London. The result, already noted, was a series of volumes on *Life and Labour of the People of London*, published in seventeen volumes between 1891 and 1903. Among many other things, this work was used to argue for the introduction of old age pensions to lessen the burdens of poverty on the elderly. Booth was by no means a "socialist" and in fact argued for reforms such as old age pensions as a means of preventing socialist revolution from coming to England. But he was clearly sympathetic to the plight of the British working classes. Part of his research methodology involved staying with working class families and recording his thoughts, findings, and observations in diaries, or as we would say today, his field notes. So in important respects he was also the first ethnographer of poverty.

The scope of Booth's work rivals anything that has been produced, say, out of the Panel Study of Income Dynamics. The work was organized into three major lines of inquiry: wages and poverty, work and industry, and religious and moral influences. The research on work and industry investigated virtually every trade and occupation in London to establish wage levels, conditions of work and employment, and relative poverty rates. The work also investigated the conditions of the unemployed—the "unoccupied classes"—and inmates of various institutions and from these aspects of the study came the first available descriptive materials on conditions of the workhouses and the causes of chronic pauperism.

Probably the best-known products of Booth's work were his famous poverty maps that showed social conditions of every street in London as of 1889. These maps can still be downloaded from the web site of the London School of Economics and through websites at the University of Michigan. They are pioneering efforts in social geography and in the study of the spatial aspects of social inequality, a line of inquiry that GIS data bases has recently resurrected in U.S. sociology.

Booth soldiered on through his studies of London poverty for two decades, evidently "cursing every minute I gave to it" as Booth said in a letter to Alfred Marshall in 1894. As one commentator on Booth's work has remarked:

Working without a thesis to prove or disprove and with such an infinity of potential subject matter to describe, it must have been difficult to know when to stop. In his introduction to [one of the seventeen

volumes], Booth makes a wry reference to the internal momentum of the inquiry: "I have at times doubted whether the prolongation of this work has had any other basis than an inability on my part to come to a conclusion."⁶

By the time sociologists of my generation got to graduate school (in the late 1960s and early 1970s), people like Galton, Quetelet, and Booth had entirely disappeared from all discussions of the origins of the discipline. So far as the founders of sociology were concerned, we learned about Marx, Weber, Durkheim, and perhaps Tonnies, Simmel, and Spencer, but nothing about the political arithmeticians, or about Marx and Weber's abortive efforts at real empirical work (see Wright 1988, for a discussion) or about Booth's landmark poverty studies in London. In fact, we were taught practically nothing at all about what Anthony Oberschall (1972) referred to as "the establishment of empirical sociology."

In fact, to the extent we were taught much of anything about the origins of sociology of the sort we were clearly being trained to practice, it was that 20th century American sociology had evolved through three eras or schools: the "Chicago School" that reigned supreme into the 1930s, the "Columbia School" of the 1940s and 1950s, and the then-nascent "Michigan School," the dominant intellectual force in much of American social science (sociology, social psychology, political science, and behavioral economics) for the past half-century. And even here, the applied or practical origins of each of these schools were not stressed as a dominant theme, or for that matter, even mentioned. Again, to the extent that the history of the discipline was taught at all, it was the version promulgated by Talcott Parsons in *The Social System* (1951) and *The Structure of Social Action* (1937), which, of course, is dominated by exegeses on Weber, Marshall, Pareto, Freud, and Durkheim, and neither of which contains so much as a single reference to any of the people and works discussed in this paper.⁷

So lest anyone be misled into thinking that the Chicago, Columbia, and Michigan "schools" of sociology were founded with the idea of testing hypotheses derived from Weber, Marshall, Pareto, Durkheim, Freud, or Parsons, let me comment briefly on the applied interests from whence all three schools were conceived.

All the early classic Chicago work—Burgess and Park's work on concentric zones as a metaphor for urban development, Thrasher's study of juvenile gangs in Chicago, Anderson's studies of hoboos, Reckless's study of bordellos, Sutherland's studies of crime—were animated specifically by social policy and reformist concerns. Some of the Chicago work was ethnographic, some more quantitative, but all of it was directed to the very practical question of whether urbanization itself was a, or perhaps the, cause of contemporary social problems. W. I. Thomas and Florian Znaniecki were later rendered as prominent social theorists, as were many of the other early Chicago sociologists, but their initial empirical work was focused on the question whether the ecological distribution of social problems such as alcoholism, homicide, suicide, mental illness, and poverty could be ascribed to the sudden freedom of immigrants released from the social controls of Europe. In all respects, the Chicago school, especially in its formative years, was populated by practical people eager to solve social ills by applying their theories, concepts, and methods in an urban laboratory. The Chicago School was applied social research.

As for the Columbia School, its founding eminence was Paul Lazarsfeld.⁸ Lazarsfeld, of course, was a European emigre who settled in Columbia University in the late 1930's. His later-to-be-renowned research shop was established as the Office of Radio Research and conducted applied surveys of the radio audience for this newly developing mass media. Much of Lazarsfeld's contribution to basic social science, particularly his work on interpersonal influence, grew directly from the findings of the radio research program. Then later, when the mission of the Office of Radio Research

became too restrictive to accommodate Lazarsfeld's expansive research interests, it was reformed and renamed as the Bureau of Applied Social Research, from whose early products, especially the voting surveys, emanated the so-called Columbia School, which came ultimately to dominate empirical social science for the next two decades. Many innovations in survey design and analysis, among them the panel survey and latent structure analysis, originated in the *applied* research undertaken by Lazarsfeld and his associates in this period. Surely, their most influential work was the voting research: the study of the 1940 presidential election that culminated in *The People's Choice* (1944) and the even more influential study of the 1948 election that gave us *Voting* (1954).

When the Survey Research Center at Michigan picked up the study of mass electoral behavior, Michigan became the dominant "school" in American sociology. Here the key figure was Angus Campbell. Campbell learned the survey research method as a staff scientist in the Department of Agriculture's Division of Program Surveys during World War II. When that division was abolished by Congressional mandate (an action initiated by a coalition of conservative Southern legislators who objected to a number of community surveys done by the division showing the continued oppression and misery of Southern blacks), Campbell, along with fellow staff members Rensis Likert and Leslie Kish, moved en masse to the University of Michigan to found the Survey Research Center, which has since become the premier survey research organization in the world. Thus even the preeminent Michigan Survey Research Center had its origins in USDA program surveys. And when Campbell teamed up with Phillip Converse, Warren Miller, and Donald Stokes to produce *The American Voter* (1960), their study of the 1956 presidential election, the transfer of intellectual influence from Columbia to Michigan was complete.

Peter Rossi (1987) has written of his experiences as a sociology student in the 1930s:

I soon learned from observing my instructors that sociological research was primarily a field in which scholars read about the world (mostly in the *New York Times*), read what other scholarly sociologists wrote about the world, and then wrote their own commentaries about what they had read. . . . To be sure, there was some empirical research going on back in those days, but most of that sort of work was kept carefully away from the undergraduate curriculum. . . . Indeed, what little I learned about empirical research was from courses taken in the Department of Psychology.

The point is that the real founders of empirical sociology as it is currently practiced were all motivated by applied, practical, reformist concerns, concerns of the sort that these days we call "applied sociology." Applied sociology, that is, is not some inferior derivative of the more noble "basic" science but lies at the very heart and core of the discipline's origins and historical enterprise. Elesh (1972) pointed out many years ago that the various statistical societies in Britain and on the Continent that came into being in the mid-to-late 19th century—the Manchester Statistical Society (1833),⁹ the Royal Statistical Society (1834) (founded, incidentally, by Quetelet), the International Statistics Congress (1853), and even the American Statistical Association (1839)—were all founded well before the British or American Sociological Associations came into being (the ASA was founded in 1905, the BSA not until 1951) and even before Durkheim's *L'Année Sociologique* (1898), but were sponsoring or in many cases doing themselves research work that today we can easily recognize as precursors to sociology as actually practiced. So it is appropriate to conclude with a passage from an address delivered to the Annual Meeting of the Royal Statistical Society by Sir Stafford Northcote in 1869:

The statistician [or, as we would say today, the sociologist] is not animated by a mere spirit of curiosity, nor does he content himself with the simple documentation of facts. His objects are at once nobler and

more practical. He aims at discovering the actual condition of his country and the causes of that condition, with a view to discover also the methods of improving it. (Cole 1972)

Sociologists engaged in applied work will do well to remember Sir Northcote as having expressed, more than a century ago, our true historical mission.

NOTES

1. The brief information about the conventional founding fathers, including birth and death dates, comes from the relevant entries in George Ritzer's *Blackwell Encyclopedia of Sociology Online* <http://www.sociologyencyclopedia.com/public/>.
2. Weber's reputation as an abstract sociological theorist is largely due to the discussion of his work in Parsons (1937, Vol. II), the collection of Weber's essays published by Gerth and Mills (1946), and by many dozens of secondary discussions that rely more on Parsons' account than on anything written by Weber himself.
3. The sources for most of my discussion of Galton are those found at <http://www.galton.org/> and at <http://www.indiana.edu/~intell/galton.shtml>. The authors of the essays to be found at these Web sites are unspecified so I am unable to acknowledge them by name.
4. Again, unattributed essays on the following web pages provide a great deal of the material summarized in this account: <http://www.morris.umn.edu/~sungurea/introstat/history/w98/Quetelet.html> and <http://www.umass.edu/wsp/statistics/tales/quetelet.html>.
5. Source: The unattributed essay at <http://www.umass.edu/wsp/statistics/tales/quetelet.html>.
6. Source: The excellent but unattributed essay on Booth at <http://booth.lse.ac.uk/static/a/2.html>.
7. In *The Social System*, Parsons speaks of "the great founders of modern social science, of whom . . . Pareto, Durkheim, and Max Weber stand out, and in addition to them especially Freud" (1951: xi).
8. Information about Lazarsfeld and the Columbia School, and the later discussion of Angus Campbell and the founding of the Michigan Survey Research Center, are from Rossi, Wright, and Anderson, 1983.
9. The Web site of the Manchester Statistical Society boasts, correctly, that it "was the first organization in Britain to study social problems systematically and to collect statistics for social purposes."

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